FORGE FITNESS

Adriaan van der Westhuizen 240436415

T2_Individual

Contents

Business Rules: 3
ERD Diagrams: 6
OVERALL ERD 8
Reflection: 9
Wireframes: 10

Business Rules:

FORGE FITNESS

Business Rules

1.User Authentication and Registration :

1.1 A new member should provide their ID for verification and create a password.

1.2 Password must meet the systems security requirements (min. length characters).

1.3 A user may only get access to member-only features when their membership is active and verified.

1.4 The system should verify login credentials before giving access.

1.5 User are only allowed to edit their own profile they are logged in with.

2.Membership :

2.1 User must be a registered and verified member of the gym to access membership.

2.2 Memberships are provided in various ways depending on the type of member.(Student , normal citizen, or elderly).

2.3 Memberships automatically expire at the end of the package duration.

2.4 Members may upgrade their package at any time, however downgrading is only permitted at the end of the current billing cycle.

2.5 A member may terminate their membership through their profile settings ; their data will be archived into the database and access will be revoked.

3.Packages :

3.1 Only verified and registered members can buy packages.

3.2 Each package must have a defined price , description and duration .

3.3 A member can only have one package at a time (active package).

4.Courses

4.1 Course content should be evaluated before being published.

4.2 Each course should have a title and description .

5. Personal Trainers Information :

5.1 Trainer profile should be provided with a name , qualification and background.

5.2 Only registered trainers may edit their own profile.

5.3 Trainers profiles are visible to all users. (registered or not registered)

5.4 Each trainer has a maximum capacity of assigned members to ensure quality of service.

5.5 Trainers and members can communicate only through the system's internal messaging feature to ensure privacy and record – keeping.

6. Payments and future billings :

6.1 Users pay a registration fee when signing up.

6.2 All package purchases must be recorded in the billing system.

6.3 Users should receive a email when registration was successful.

6.4 Admin may generate and review billing reports.

7. Admin Controls :

7.1 Only administrators are allowed to add and update or even delete members accounts.

7.2 Only administrators may update packages , courses or trainers details.

7.3 Administrators have enhanced privileges on their accounts.

7.4 All changes made by administrators must be logged for audit purposes.

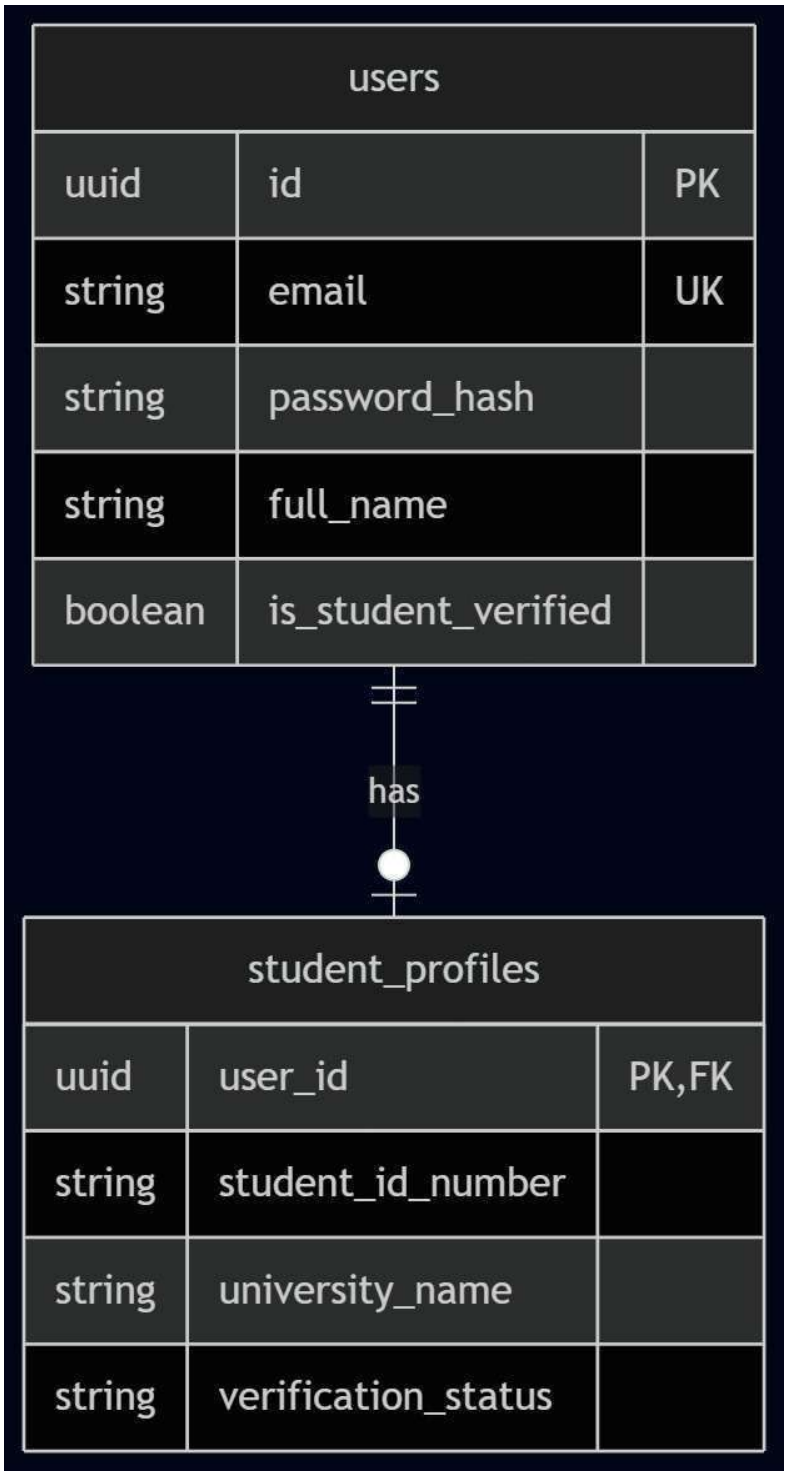
8. System Management

8.1 The System must store all user and membership information in the database.

8.2 The System must back up data regular to prevent risk of losing all data.

ERD Diagrams:

Links to every other module. Users must be logged in to enroll in courses, track plan progress, make transactions, and leave reviews.



Links to Trainers (Module 3). Each trainer is assigned to a location. Locations are displayed on the homepage and about us page. Relation: One to many. An enquirie can have only one location but a location can have many enquiries.

locations		
uuid	id	PK
string	name	
string	address	
string	hours	
string	contact_info	

contact_enquiries		
uuid	id	PK
string	name	
string	email	
text	message	
string	status	

MODULE 1 — Auth & Identity

users — every registered account on the site. Central table that almost everything else links back to since most actions require being logged in.

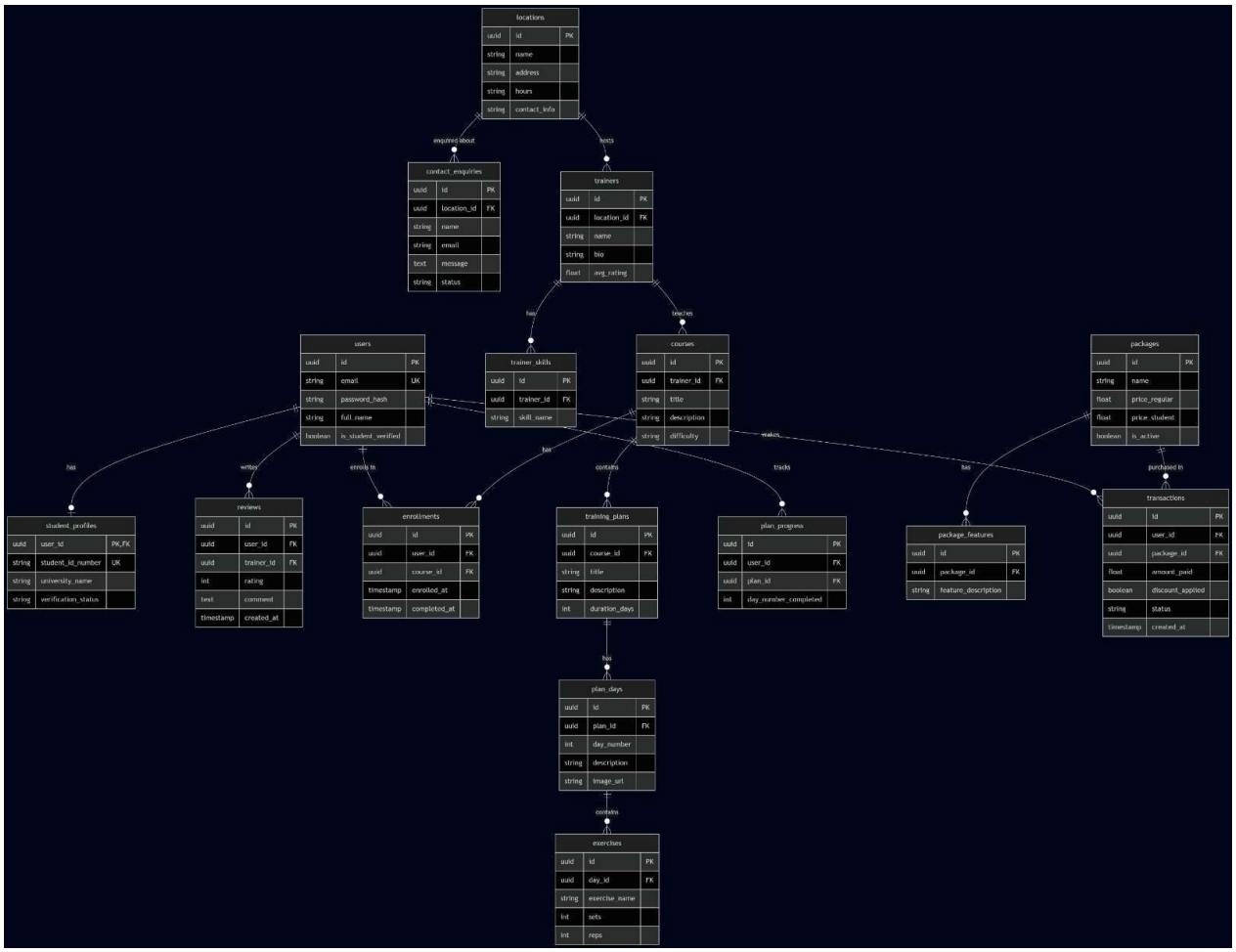
student_profiles — optional extension of users, only exists if the user is a student. Stores student number and university for discount verification purposes.

MODULE 2 — Locations & Contact

locations — each physical gym branch. Feeds into the homepage and about us page showing where the gyms are, and ties into trainers since each trainer is based at a branch.

contact enquiries — records every contact form submission. Stores the person's details and message, with a status field so the gym can track whether it's been responded to.

OVERALL ERD


locations
uuid id PK
string name
string address
string hours
string contact_info
enquires about
hosts
contact_enquiries
uuid id PK
uuid location_id FK
string name
string email
text message
string status
trainers
uuid id PK
uuid location_id FK
string name
string bio
float avg_rating
has
teaches
users
uuid id PK
string email UK
string password_hash
string full_name
boolean is_student_verified
trainer_skills
uuid id PK
uuid trainer_id FK
string skill_name
courses
uuid id PK
uuid trainer_id FK
string title
string description
string difficulty
packages
uuid id PK
string name
float price_regular
float price_student
boolean is_active
makes
has
writes
enrolls in
has
contains
tracks
has
purchased in
student_profiles
uuid user_id PK,FK
string student_id_number UK
string university_name
string verification_status
reviews
uuid id PK
uuid user_id FK
uuid trainer_id FK
int rating
text comment
timestamp created_at
enrolments
uuid id PK
uuid user_id FK
uuid course_id FK
timestamp enrolled_at
timestamp completed_at
training_plans
uuid id PK
uuid course_id FK
string title
string description
int duration_days
plan_progress
uuid id PK
uuid user_id FK
uuid plan_id FK
int day_number_completed
package_features
uuid id PK
uuid package_id FK
string feature_description
transactions
uuid id PK
uuid user_id FK
uuid package_id FK
float amount_paid
boolean discount_applied
string status
timestamp created_at
has
plan_days
uuid id PK
uuid plan_id FK
int day_number
string description
string image_url
contains
exercises
uuid id PK
uuid day_id FK
string exercise_name
int sets
int reps

Reflection:

As my role of being group leader it has been good to lead the team so far and we made good progress . So far we have made the ERD and the business rules were set up. We created our base wireframes on Figma .

Raul created the GitHub repositorie which we all forked and made branches of our own. On each branch we submitted the file StaggeredSubs_T2_INV_240436415.

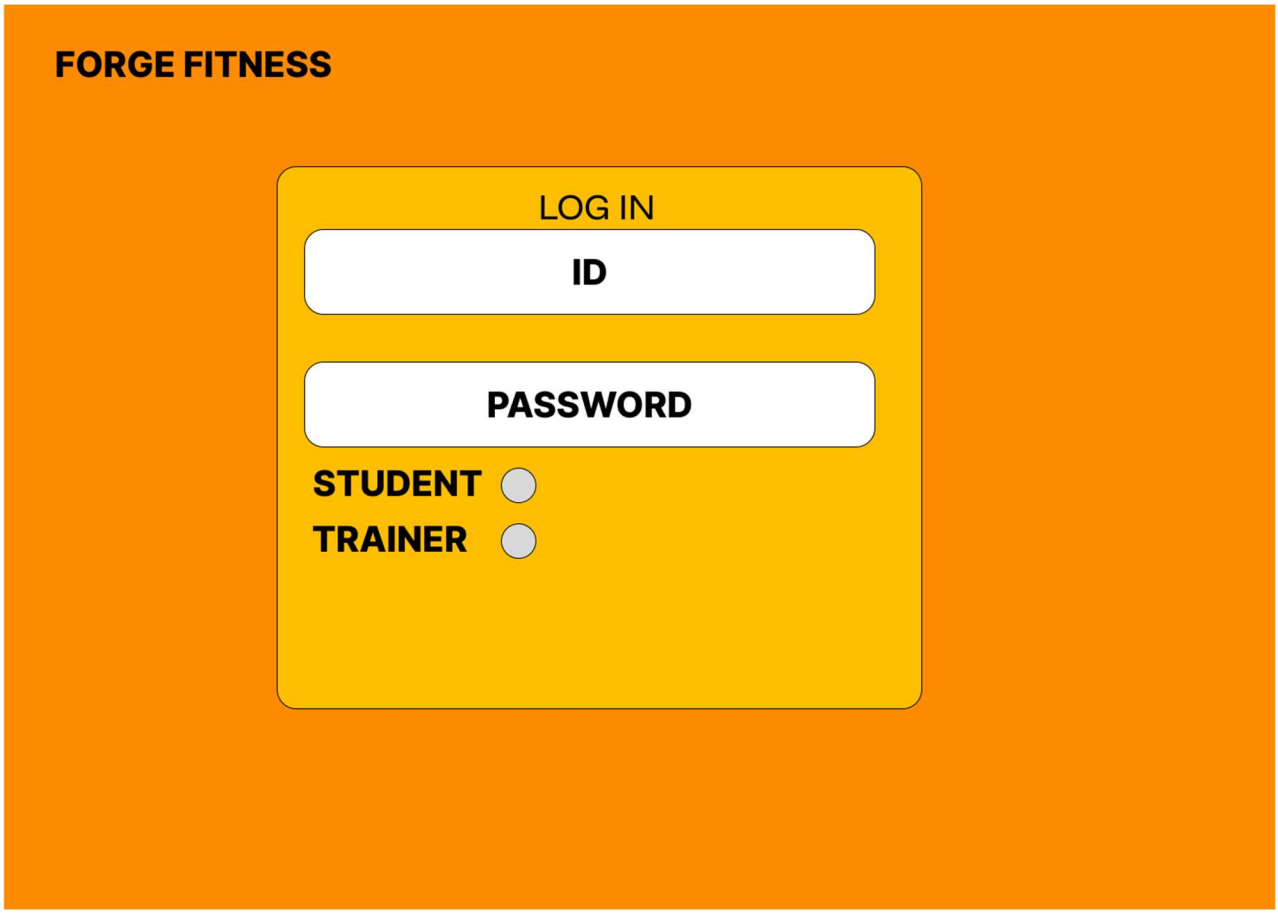
As part of the Business Rules I was responsible for the User Authentication and Registration and the System Management Sections.

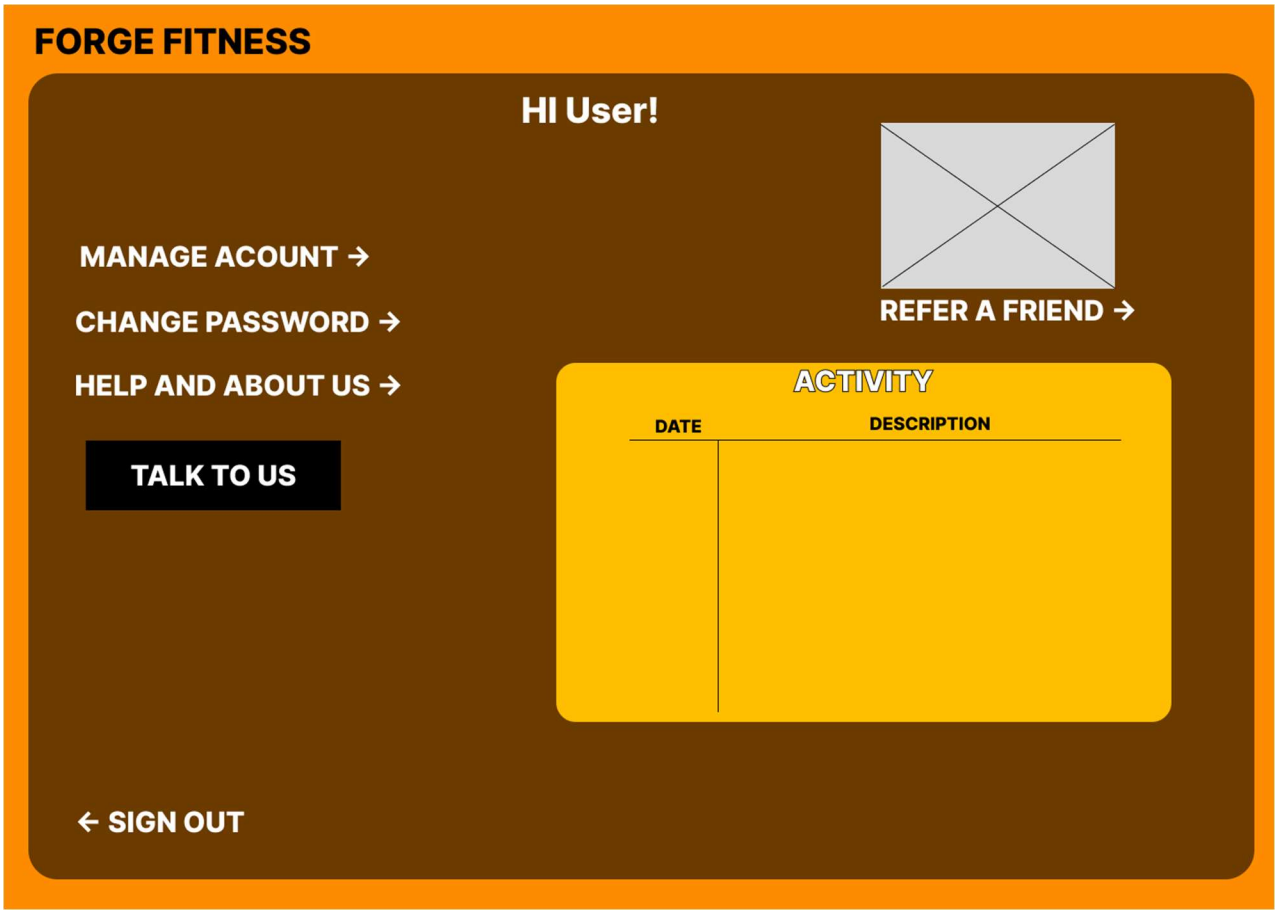
For the wireframes I was responsible for the Login , sign-up and user profile pages. I am also partnered with Dayaan to do the Home page and the about us pages.

Provided is the Figma Link :
https://www.figma.com/design/E8tdpsdQxe4oUiEYQrLmiD/ForgeFitness-Wireframes?node-id=0-1&t=2vuM2COzT93j0kuE-1

Wireframes:

LOG-IN Page Wireframe



SIGN-UP Page Wireframe

User-Profile Page Wireframe